Edoardo (Edo) Giusto

ViCAS Fellow @ UniVie · Ph.D. in Computer and Systems Engineering · Qiskit Advocate · Next Generation Internet (NGI) Ambassador

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GoogleScholar, H-index: 15 | Scopus *profile link*, H-index: 11 | ResearcherID: AAH-3056-2019 | ResearchGate | 0000-0001-8371-6685

Research Interests

Quantum Computing (QC) offers the potential to enhance traditional High-Performance Computing (HPC) workloads by leveraging the unique properties of qubits, leading to the emergence of a new computing paradigm: QHPC. My research revolves around the *dependability* of such hybrid systems, from the point of view of *reproducibility* of computation results, *resilience* to hardware and software failures, *security and privacy* of code and data.

Education

Superconducting Quantum Materials and Systems Center (SQMS) - Fermilab — *Batavia, IL, US*
US Quantum Information Summer School (USQIS) — *Aug. 7-15 2023*

Politecnico di Torino - Dept. of Control and Computer Engineering — *Torino, IT*
Ph.D. in Computer and Systems Engineering — *Nov. 1st, 2017 - Sept. 14th, 2021*
Thesis: *Sensor-based ICT systems for Smart Societies.* Defining guidelines for the correct and scientifically-sound design of a device operating in the IoT domain. Tasks taken into account: data acquisition, data transmission, and data authenticity and safe-storage.
Advisor: *Prof. Maurizio Rebaudengo.* Committee: *Antonio Liotta, Julio Perez, Luca Sterpone, Giovanni Pau, Enrico Natalizio.*

Eidgenössische Technische Hochschule (ETH) Zürich — *Zurich, CH*
Quantum Information for Developers - Summer school — *Sept. 8-12, 2019*

MITxPRO Professional Certificate Program — *MIT - Online Course*
Applications of Quantum Computing — *Apr. 2018 - Oct. 2018*

Politecnico di Torino — *Torino, IT & Paris, FR*
Master's Degree in Computer Engineering — *Sept. 1st, 2014 - July 27th, 2017*
Design and implementation of a low-cost, high-precision air pollution monitoring IoT device. Master's thesis project carried out at LiP6 - Universitè Pierre et Marie Curie, Paris, France.
Advisors: *Prof. Maurizio Rebaudengo* in Politecnico di Torino and by *Prof. Giovanni Pau* in UPMC, Paris.

Politecnico di Torino — *Torino, IT*
Bachelor's Degree in Computer Engineering — *Sept. 1st, 2010 - Mar. 31st, 2015*

Experience

University of Vienna, Vienna Center for Advanced Studies — *Vienna, Austria*
ViCAS Fellow — *Oct. 1st, 2026 - Present*

- Selected for the inaugural ViCAS fellowship cohort. Project: *Engineering Dependable Quantum High-Performance Computing Infrastructures for Future Scientific and Societal Applications.*

Helix 42 — *Rome, Italy*
Co-Founder — *March 2025 - Present*

- Head of Quantum Computing at Helix 42 *https://helix-ft.com/*.

University of Naples, Federico II - Department of Electrical Engineering and Information Technology — *Naples, Italy*
Assistant Professor - RTD-A — *Dec. 29th, 2023 - Sept. 30th, 2026*

- Position supported by the National Centre for HPC, Big Data and Quantum Computing (CN1), within the Italian "Piano Nazionale di Ripresa e Resilienza (PNRR)", Mission 4 Component 2 Investment 1.4 funded by the European Union - NextGenerationEU - CN00000013, Spoke 9 - Digital Society & Smart Cities.

Oak Ridge National Laboratory *Oak Ridge, TN, US*
Visiting Scholar *Jul. 29th, 2024 - Aug. 27th, 2024*

- Visiting period at the Quantum Science Center, hosted by *Dr. Travis S. Humble.*

University of Illinois, Urbana-Champaign *Urbana-Champaign, IL, US*
Visiting PostDoc *Nov. 27 - Dec. 1 2023*

- Visiting period in the DEPEND research group at the *Coordinated Science Laboratory*, hosted by *Prof. Ravishankar K. Iyer.*

SQMS - Fermi National Accelerator Laboratory *Batavia, IL, US*
NGI Enrichers - Fellow *Aug. 2023 - Dec. 2023*

- *CRIT-Q - Cosmic ray impact on transmon qubits*: Characterization of transient faults impact caused by cosmic rays on superconducting quantum devices. Expedition made possible by the European Union *NGI Enrichers fellowhip.* Supervisor: *Dr. Silvia Zorzetti.*

Politecnico di Torino - Department of Control and Computer Engineering *Torino, Italy*
Postdoctoral Research Assistant *Apr. 2022 - Dec. 2023*

- Reliability assessment of Quantum Computing devices, compilation and problem mapping, applications of Quantum Computing for IoT, Industry 4.0 and Financial domains.
- Investigation on Wireless Sensor Network applications for low-cost air pollution monitoring tasks. Supervisor: *Prof. Maurizio Rebaudengo.*

Funded Projects

Quantum Computing for Power Systems *Naples, IT*
Funding by: Regional Center for Information Communication Technology (CeRICT). *Nov. 2025 - Nov. 2026*
Study of the applicability of quantum computing in the electro-energy sector. Amount: ≈US $ 23000. Role: Principal Investigator.

Quantum Computing for Financial Industry *Torino, IT*
Funding by: Intesa San Paolo S.p.A. *Oct. 2021 - Dec. 2023*
Improvement of a quantum algorithm for Credit Risk Analysis. Project conducted in collaboration with *IBM Italy*. Amount: undisclosed. Role: participant.

Quantum Computing for Telecommunications Industry *Torino, IT*
Funding by: TIM S.p.A. *June 2019 - Dec. 2023*
Study of possible applications of Quantum Computing in the Telecommunications domain and development of dedicated Quantum Algorithms. Amount: ≈ US $30'000/year. Role: main developer.

Quantum Computing for Automotive Industry *Torino, IT*
Funding by: General Motors *2018 - 2020*
Study of applications of Quantum Computing in the automotive sector. Amount: ≈ US $25'000. Role: participant.

Project Proposals Under Evaluation

CRUQS – Cross-layer Reliability through a Unified fault science for Quantum-Classical Systems
Funding by: European Research Council (ERC) – ERC Starting Grant, call ERC-2027-STG (HORIZON-ERC). Proposal ID: 101398579. *Submitted Oct. 2026*
Builds a unified science of faults, errors and failures for quantum-HPC (QHPC) systems. Controlled faults are injected at five origins (correlated device disturbances, error-correction decoder faults, resource-estimation defects, compiler/mapping faults, scheduling/orchestration faults) and traced across quantum-classical interfaces. The goal is validated reliability principles that hold across qubit technologies, show when local protection is enough and when dependable QHPC needs action across the quantum-classical boundary, and give a scientific basis for future standards. Duration: 60 months. Team: PI, 2 postdocs, 2 PhD students. Amount: EUR 1,492,879 requested EU contribution. Role: Principal Investigator.

Q-MAESTRO – Quantum Benchmarking Orchestration Framework for Europe
Funding by: Horizon Europe – EuroHPC Joint Undertaking, call HORIZON-EUROHPC-JU-2024-BENCHMARK-05. Proposal no.: 101336056.
Consortium project on a European framework for benchmarking quantum computers integrated in HPC. The University of Naples Federico II (UniNa) unit contributes to WP5 with integration-centric KPIs, statistical robustness analysis for hybrid quantum-classical pipelines, and dependability and resilience experiments that measure how noise affects the statistical stability of benchmark results. Duration: 36 months. Amount (UniNa unit): EUR 36,549 requested EU contribution (EUR 73,099 total eligible cost; 5 person-months). Role: Leading researcher for the UniNa unit.

Honors & Awards

2026	**ACM DAC 2026 Chips to Systems Conference**. Outstanding TPC Member	*Award*
2023	**NGI - Next Generation Internet Enrichers - Transatlantic Fellowship**. Funding to carry out a 5-months visiting research period at SQMS - Fermilab. Amount: ≈ US $25'000 .	*Fellowship*
2021	**Politecnico di Torino**. 3rd Year PhD Award Competition 2021. Amount: ≈ US $1'500.	*3rd Prize*
2020	**Politecnico di Torino**. 2nd Year PhD Award Competition 2020. Amount: ≈ US $1'500.	*3rd Prize*
2020	**IBM Quantum**. Challenge 2020 Achievement.	*IBM Quantum Challenge Badge*
2019	**IBM Quantum**. Qiskit Advocate Badge.	*Qiskit Advocate Badge*
2019	**IBM Qiskit Camp Europe 2019 - Quantum Computing Hackathon**. Winning team member, *"Quantum Synth: a quantum-computer-based music synthesizer"*. Team composed by: Costa Hamido, O.; Ghazi Vakili, M.; Giusto, E.; Baiardi, A.; and Cirillo, G.A. Video description available *here*. Related conference publication available *here*.	*Community Choice Award*

Top 5 Publications

Complete list at 🎓*here*

Dependable Classical-Quantum Computing Systems Engineering
Frontiers in Computer Science (Hypothesis and Theory article, Open Access) *2025*
E. Giusto, S. Núñez-Corrales, K. N. Smith, P. Cao, E. Younis, P. Rech, F. Vella, B. Baheri, A. Cilardo, B. Montrucchio, W. Jiang, S. Xu, S. Dasgupta, R. K. Iyer, T. S. Humble; https://doi.org/10.3389/fcomp.2025.1520903. Citations: 10. Keywords: hybrid classical-quantum systems, HPC, quantum computing, dependability, reliability, resiliency, security, reproducibility.

- *Venue*: Frontiers in Computer Science is an open-access journal reaching the HPC, quantum computing and security communities. As first author and scientific lead, I coordinated 15 authors from 12 institutions, including ORNL, LBNL and NCSA/UIUC.
- *Impact*: This paper proposes the first framework for dependable quantum-HPC (QHPC) systems, built on three pillars (reproducibility, resiliency, security & privacy), and defines six engineering principles of dependability with an R&D roadmap for QC–HPC integration.
- *Research Directions*: It sets dependability as a first-class design goal for QHPC, calling for reproducibility tolerances for drifting devices, quantum resilience metrics, a trusted quantum computing base, and an open quantum-classical OS and compiler.

A Typology of Quantum-Classical Faults *Naples, IT*
IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W 2025), FORCE Workshop *June 2025*
E. Giusto, S. Núñez-Corrales, A. Cilardo, N. Mazzocca, T. S. Humble; https://doi.org/10.1109/DSN-W65791.2025.00060. Citations: 3. Keywords: quantum computing, quantum system engineering, QHPC, dependability, hybrid quantum-classical systems.

- *Venue*: Published at DSN, the flagship dependability conference, as the founding paper of the FORCE workshop, in collaboration with the Quantum Science Center at ORNL and NCSA/UIUC.
- *Impact*: This paper extends the classical Avižienis et al. fault taxonomy to hybrid quantum-classical systems, adding quantum-specific faults, and shows how the integration pattern (tight, on-premise, cloud) and qubit technology shape how faults appear and spread.
- *Research Directions*: It gives a shared vocabulary for quantum-classical faults and argues for actionable fault profiles, cross-layer observability and fault-aware resource management. These ideas underpin my ERC Starting Grant proposal CRUQS.

Understanding Logical-Shift Error Propagation in Quanvolutional Neural Networks
IEEE Transactions on Quantum Engineering (Impact Factor: 4.4, SJR: Q1.) *2024*
M. Vallero, E. Dri, **E. Giusto**, B. Montrucchio and P. Rech; https://doi.org/10.1109/TQE.2024.3372880. Citations: 10. Keywords: Fault injection, fault tolerance, quantum computing (QC), quantum machine learning (QML), reliability evaluation.

- *Venue*: IEEE Transactions on Quantum Engineering (TQE) is among the few leading journals focusing exclusively on quantum engineering, making it the ideal venue to publish this cross-disciplinary work between quantum machine learning and fault tolerance.
- *Impact*: This is the first study to characterize the logical-shift error propagation in Quanvolutional Neural Networks (QNNs), injecting over 13 billion faults. It shows that even small phase shifts in the quanvolutional layer can cause misclassifications, and identifies fault-sensitive substructures (e.g., control vs. target qubits, dataset-specific vulnerabilities).
- *Research Directions*: This paper pioneers fault-aware quantum ML design, emphasizing the need for resilient quantum ML architectures. It invites further research into application-specific fault models, dynamic error mitigation for QNNs, and joint optimization of accuracy and reliability in hybrid quantum-classical inference pipeline.

A Systematic Methodology to Compute the Quantum Vulnerability Factors for Quantum Circuits
IEEE Transactions on Dependable and Secure Computing (Impact Factor: 7.3, SJR: Q1) *2023*
D. Oliveira, **E. Giusto**, B. Baheri, Q. Guan, B. Montrucchio, P. Rech; https://doi.org/10.1109/TDSC.2023.3313934 - Citations: 32. Keywords: Fault injection, QVF metric, quantum computing, reliability evaluation.

- *Venue*: IEEE Transactions on Dependable and Secure Computing (TDSC) is one of the premier journals in the field of dependable systems, making this publication a significant milestone in positioning dependability as a first-class concern in quantum computing.
- *Impact*: This paper formalizes the notion of Quantum Vulnerability Factor (QVF), drawing from classical AVF/PVF metrics, to quantify how susceptible a quantum circuit is to transient faults. Through more than 388 million injections across diverse fault types, the study provides a robust, empirical basis for understanding how phase shifts affect circuit reliability.
- *Research Directions*: The proposed QVF metric opens a path for quantum-aware design automation, fault-aware qubit mapping strategies, and the development of quantum reliability compilers. It also lays the groundwork for extending fault modeling beyond noise-centric abstractions—crucial as we scale NISQ devices and integrate them into hybrid HPC-QC systems.

QuFI: a Quantum Fault Injector to Measure the Reliability of Qubits and Quantum Circuits *Baltimore, MD, US*

IEEE/IFIP International Conference on Dependable Systems and Networks (DSN22) (Acceptance rate: 18%, Core Ranking: A) *June 27-30, 2022*

D. Oliveira, **E. Giusto**, E. Dri, N. Casciola, B. Baheri, Q. Guan, B. Montrucchio, P. Rech; https://doi.org/10.1109/DSN53405.2022.00025. Citations: 30. Keywords: Quantum Computing, Fault Tolerance, Fault Injection. Reliability Evaluation.

- *Venue*: IEEE/IFIP International Conference on Dependable Systems and Networks (DSN) is the flagship event in the dependability community. A presentation here underlines the significance of bridging traditional fault tolerance with the realities of quantum computation.
- *Impact*: This paper introduces QuFI, the first publicly available quantum fault injector supporting both simulation and real-machine fault injection on IBM Q. It models radiation-induced transient faults as parametric phase shifts, capturing non-binary fault behavior—a key departure from classical bit flips. It quantifies the correlation between fault injection, qubit locality, and output corruption.
- *Research Directions*: This foundational tool enables systematic fault modeling for radiation-aware quantum architectures, facilitates comparative studies of fault tolerance across hardware generations, and promotes open benchmarking of error sensitivity across algorithms and workloads—an essential capability as reproducibility and resilience become interlinked pillars of dependable QHPC systems.

Publications - Journal papers

Complete list at 🎓*here*

Evolutionary tailoring of quantum kernels for cyber-physical anomaly detection: A case study on the SWaT testbed
Applied Soft Computing (SJR: Q1) *2026*

G. Acampora, E. Giusto, A. Senese, N. Mazzocca; https://doi.org/10.1016/j.asoc.2026.115664.

Dependable classical-quantum computing systems engineering
Frontiers in Computer Science *2025*

E. Giusto, S. Núñez-Corrales, K. N. Smith, P. Cao, E. Younis, P. Rech, F. Vella, B. Baheri, A. Cilardo, B. Montrucchio, W. Jiang, S. Xu, S. Dasgupta, R. K. Iyer, T. S. Humble; https://doi.org/10.3389/fcomp.2025.1520903. Citations: 10.

Understanding Logical-Shift Error Propagation in Quanvolutional Neural Networks
IEEE Transactions on Quantum Engineering (Impact Factor: 4.4, SJR: Q1.) *2024*

M. Vallero, E. Dri, E. Giusto, B. Montrucchio and P. Rech; https://doi.org/10.1109/TQE.2024.3372880. Citations: 10.

Quantum Computing in Finance: The Intesa Sanpaolo Experience
IEEE Engineering Management Review (SJR: Q2) *2024*

R. Sotelo, D. Corbelletto, E. Dri, E. Giusto and B. Montrucchio; https://doi.org/10.1109/EMR.2024.3373796. Citations: 14.

A Systematic Methodology to Compute the Quantum Vulnerability Factors for Quantum Circuits
IEEE Transactions on Dependable and Secure Computing (Impact Factor: 7.3, SJR: Q1) *2023*

D. Oliveira, E. Giusto, B. Baheri, Q. Guan, B. Montrucchio, P. Rech; https://doi.org/10.1109/TDSC.2023.3313934. Citations: 32

A More General Quantum Credit Risk Analysis Framework
MDPI Entropy (Impact Factor: 2.7, SJR: Q2) *2023*

E. Dri, A. Aita, E. Giusto, D. Ricossa, D. Corbelletto, B. Montrucchio, R. Ugoccioni; Volume 25, number 4, 2023. https://doi.org/10.3390/e25040593. Citations: 28.

Air-to-Ground Transmission and Near Real-Time Visualization of FBG Sensor Data via Cloud Database
IEEE Sensors Journal (Impact Factor: 4.3, SJR: Q1) *2022*

A. Marceddu, G. Quattrocchi, A. Aimasso, E. Giusto, L. Baldo, M. Ghazi Vakili, M. D. L. Dalla Vedova, B. Montrucchio, P. Maggiore; https://doi.org/10.1109/JSEN.2022.3227463. Citations: 19.

A fuzzy control system for energy-efficient wireless devices in the Internet of vehicles
Wiley International Journal of Intelligent Systems (Impact Factor: 7, Acceptance Rate: 12%, SJR: Q1) *2021*

M. Collotta, R. Ferrero, E, Giusto, M. Ghazi Vakili, J. Grecuccio, X Kong, I. You; https://doi.org/10.1002/int.22353. Citations: 23.

A Densely-Deployed, High Sampling Rate, Open-Source Air Pollution Monitoring WSN
IEEE Transactions on Vehicular Technologies (Impact factor: 6.8, SJR: Q1) *2020*

B. Montrucchio, E. Giusto, M. Ghazi Vakili, S. Quer, C. Fornaro; Volume 69, Issue 12. https://doi.org/10.1109/TVT.2020.3035554. Citations: 70.

Combining Blockchain and IoT: Food-Chain Traceability and Beyond
MDPI Energies (Impact Factor: 3.2, SJR: Q1) *2020*
J. Grecuccio, E. Giusto, F. Fiori, M. Rebaudengo ; *Combining Blockchain and IoT: Food-Chain Traceability and Beyond.* Volume 13 (15), 3820. https://doi.org/10.3390/en13153820. Citations: 107.

Quantum pliers cutting the Blockchain
IEEE IT Professional (Impact factor: 2.6, SJR: Q2) *2020*
E. Giusto, M. Ghazi Vakili, F. Gandino, C. Demartini, B. Montrucchio; Vol.22, Issue 6. https://doi.org/10.1109/MITP.2020.2974690. Citations: 9.

Publications - Conference Proceedings

Complete list at 🎓*here*

Productive Quantum Programming Needs Better Abstract Machines *Albuquerque, NM, US*
IEEE International Conference on Quantum Computing and Engineering (QCE25) *Aug. 31 - Sept. 5, 2025*
S. Núñez-Corrales, O. Di Matteo, J. Dumbell, M. Edwards, E. Giusto, S. Pakin, V. Stirbu. https://doi.org/10.1109/QCE65121.2025.00094. Citations: 3.

A Typology of Quantum-Classical Faults *Naples, IT*
Dependable Systems and Networks Workshops (DSN-W) *June 26, 2025*
E. Giusto, S. Núñez-Corrales, A. Cilardo, N. Mazzocca and T. Humble. https://doi.org/10.1109/DSN-W65791.2025.00060. Citations: 3.

Harnessing a 256-qubit Neutral Atom Simulator for Graph Classification *Montreal, CA*
IEEE International Conference on Quantum Computing and Engineering (QCE24) *Sept. 15–20, 2024*
E. Giusto, G. Iurlaro, B. Montrucchio, A. Scionti, O. Terzo, C. Vercellino, G. Vitali, P. Viviani. https://doi.org/10.1109/QCE60285.2024.00043.

Q-SCALE: Quantum computing-based Sensor Calibration for Advanced Learning and Efficiency *Montreal, CA*
IEEE International Conference on Quantum Computing and Engineering (QCE24) *Sept. 15–20, 2024*
L. Bergadano, A. Ceschini, P. Chiavassa, E. Giusto, B. Montrucchio, M. Panella, A. Rosato. https://doi.org/10.1109/QCE60285.2024.00044. Citations: 6.

Quantum Kernel Estimation With Neutral Atoms For Supervised Classification: A Gate-Based Approach *Bellevue, WA, US*
IEEE International Conference on Quantum Computing and Engineering (QCE23) *Sept. 17–22, 2023*
M. Russo, E. Giusto, B. Montrucchio; https://doi.org/10.1109/QCE57702.2023.00032. Citations: 8.

BBQ-mIS: a parallel quantum algorithm for graph coloring problems *Bellevue, WA, US*
WIHPQC @ IEEE International Conference on Quantum Computing and Engineering (QCE23) *Sept. 17–22, 2023*
C. Vercellino, G. Vitali, P. Viviani, E. Giusto, A. Scionti, A. Scarabosio, O. Terzo, B. Montrucchio; WIHPQC, Third International Workshop on Integrating High-Performance and Quantum Computing, co-located with QCE23. https://doi.org/10.1109/QCE57702.2023.10198. Citations: 10.

Towards An End-To-End Approach For Quantum Principal Component Analysis *Bellevue, WA, US*
QML @ IEEE International Conference on Quantum Computing and Engineering (QCE23) *Sept. 17–22, 2023*
E. Dri, A. Aita, T. Fioravanti, G. Franco, E. Giusto, G. Ranieri, D. Corbelletto, B. Montrucchio; International Workshop on Quantum Machine Learning: From Foundations to Applications, co-located with QCE23. https://doi.org/10.1109/QCE57702.2023.10175. Citations: 19.

Understanding the Effect of Transpilation in the Reliability of Quantum Circuits *Bellevue, WA, US*
StableQ @ IEEE International Conference on Quantum Computing and Engineering (QCE23) *Sept. 17–22, 2023*
N. Dilillo, E. Giusto, E. Dri, B. Baheri, Q. Guan, B. Montrucchio, P. Rech; Quantum System Stability and Reproducibility Workshop, co-located with QCE23. https://doi.org/10.1109/QCE57702.2023.10220. Citations: 2.

Neural optimization for quantum architectures: graph embedding problems with Distance Encoder Networks *Torino, IT*
IEEE International Computer Software and Applications Conference (COMPSAC) (Acceptance rate: 26%, CORE Ranking: B) *June 26-30, 2023*
C. Vercellino, G. Vitali, P. Viviani, A. Scionti, A. Scarabosio, O. Terzo, E. Giusto, B. Montrucchio; https://doi.org/10.1109/COMPSAC57700.2023.00058. Citations: 5.

Comparison of heuristic approaches to PCI planning for Quantum Computers *Las Vegas, NV, US*
IEEE International Conference on Consumer Electronics (ICCE23) *Jan. 6-8, 2023*
G. Barillaro, A. Boella, F. Gandino, M. Ghazi Vakili, E. Giusto, G. Mondo, B. Montrucchio, A. Scarabosio, A. Scionti, O. Terzo , G. Vitali ; https://doi.org/10.1109/ICCE56470.2023.10043394. Citations: 8.

Towards practical Quantum Credit Risk Analysis *Teddington, UK*
NPL Joint Symposium on Quantum Technologies *Sept. 13-14, 2022*
E. Dri, E. Giusto, A. Aita, B. Montrucchio; https://doi.org/10.1088/1742-6596/2416/1/012002. Citations: 15.

Neural-powered unit disk graph embedding: qubits connectivity for some QUBO problems *Broomfield, CO, US*
IEEE International Conference on Quantum Computing and Engineering (QCE22) *Sept. 18-23, 2022*
C. Vercellino, P. Viviani, G. Vitali, A. Scionti, A. Scarabosio, O. Terzo, E. Giusto, B. Montrucchio; https://doi.ieeecomputersociety.org/10.1109/QCE53715.2022.00038. Citations: 16.

Understanding the Impact of Cutting in Quantum Circuits Reliability to Transient Faults
2022 IEEE 28th International Symposium on On-Line Testing and Robust System Design (IOLTS) *2022*
N. Casciola, E. Giusto, E. Dri, D. Oliveira, P. Rech, B. Montrucchio; pp. 1-7. https://doi.org/10.1109/IOLTS56730.2022.9897308. Citations: 8.

QuFI: a Quantum Fault Injector to Measure the Reliability of Qubits and Quantum Circuits *Baltimore, MD, US*
IEEE/IFIP International Conference on Dependable Systems and Networks (DSN22) (Acceptance rate: 18%, Core Ranking: A) *June 27-30, 2022*
D. Oliveira, E. Giusto, E. Dri, N. Casciola, B. Baheri, Q. Guan, B. Montrucchio, P. Rech; https://doi.org/10.1109/DSN53405.2022.00025. Citations: 30.

Quantum synth: a quantum-computing-based synthesizer *Graz, AT*
ACM International Conference on Audio Mostly (AM'20) *Sept. 15-17, 2020*
O. Costa Hamido, G.A. Cirillo, E. Giusto; Pages 265–268. https://doi.org/10.1145/3411109.3411135. Citations: 22.

Research Community Engagement

Software for the Integration of Quantum and HPC Ecosystems Workshop @ SC 2026
Organizer

- Organizer for Software for the Integration of Quantum and HPC Ecosystems workshop at IEEE/ACM International Conference for High Performance Computing, Networking, Storage, and Analysis (SC26), website *here*.

IEEE Quantum Week (QCE) 2026
Member at large

- Technical Committee Member for *Quantum Applications (QAPP) track*.
- Organizing Committee Member for *Enabling Programming Abstractions for More Effective Quantum Development* Workshop, website *here*.

Foundations Of Reliable Classical-quantum Engineering (FORCE) Workshop @ DSN 2025,2026
General Chair

- Organizer for FORCE workshop at IEEE/IFIP Dependable Systems and Networks (DSN) 2025,2026, website *here*.

IEEE/ACM DAC Chips to Systems Conference (DAC) 2025, 2026, 2027
Technical Program Committee member

- Technical Program Committee member for track *DES6 - Quantum Computing*, Meta-reviewer duties, website *here*.

IEEE/ACM Design, Automation and Test in Europe (DATE) 2025, 2026, 2027
Technical Program Committee member

- Technical Program Committee member for track *D16 - Design Automation for Quantum Computing*, website *here*.

Software for the Integration of Quantum and HPC Ecosystems Workshop @ SC 2025
Organizer

- Organizer for Software for the Integration of Quantum and HPC Ecosystems workshop at IEEE/ACM International Conference for High Performance Computing, Networking, Storage, and Analysis (SC25), website *here*.

IEEE Quantum Week (QCE) 2025
Member at large

- Technical Committee Member for *Quantum Applications (QAPP) track*.
- Organizing Committee Member for *Enabling Programming Abstractions for More Effective Quantum Development* Workshop, website *here*.
- Organizing Committee Member for *4th Workshop on Quantum in Consumer Technology*, website *here*.

Software Frameworks for integrating Quantum and HPC Ecosystems Workshop @ ICS 2025
Program Committee Member

- Program Committee Member for Software Frameworks for integrating Quantum and HPC Ecosystems workshop at ACM International Conference on Supercomputing (ICS25), website *here*.

Quantum Classical Cooperative Computing (QCCC) Workshop @ ISCA 2025
Program Committee Member

- Program Committee Member for QCCC workshop at IEEE/ACM The International Symposium on Computer Architecture (ISCA), website *here*.

Special Session @ ETS 2025
Organizer

- Organizer for the Special session *Good, Bad, & Ugly in Quantum Computing: Computational Power, Intrinsic Noise & Transient Faults* at IEEE European Test Symposium (ETS), website *here*.

HPC/AI Integration with Quantum Computing (HAIQ) Workshop @ HPCA 2025
Program Committee Member

- Program Committee Member for HAIQ workshop at IEEE International Symposium on High-Performance Computer Architecture (HPCA), website *here*.

IEEE Quantum Week (QCE) 2024
Member at large

- General Chair for *1st Workshop on Dependability Challenges in Hybrid Classical-Quantum Computing Systems*, Workshop website *here*.
- Technical Committee Member for *Quantum Technologies and Systems Engineering (QTEM) track*.
- Technical Committee Member for *Quantum Applications (QAPP) track*.
- Organizing Committee Member for *3rd Workshop on Quantum in Consumer Technology*, Workshop website *here*.

UIUC - Online Workshop
General Chair

- Organizer for UIUC/NCSA-hosted workshop *Virtual Workshop on Dependable Classical-Quantum Computing Systems Engineering (DCQCS)*, website *here*.

IEEE Workshop on Quantum IntelLigence, Learning & Security (QUILLS) @ TPS 2024, 2025
Program Committee Member

- Program Committee Member for QUILLS workshop, website *here*.

Frontiers in Computer Science
Topic Editor

- Topic Editor for *Realizing Quantum Utility: Grand Challenges of Secure & Trustworthy Quantum Computing*, website *here*.

IEEE International Conference on Consumer Electronics (ICCE) 2024
Track Chair

- Track: *Quantum In Consumer Technology*.

IEEE Quantum Week (QCE) 2023
Member at large

- Technical Committee Member for *Quantum Technologies and Systems Engineering (QTEM) track*.
- Technical Committee Member for *Quantum System Stability and Reproducibility Workshop*.
- Organizing Committee Member for *2nd Workshop on Quantum in Consumer Technology*, Workshop agenda *here*.

IEEE Consumer Technology Society
Technical Committee Member

- Technical Committee Member for *Quantum in Consumer Technology*.

Peer Reviewing - Journals

- IEEE - *Transaction on Nuclear Science*
- IEEE - *Internet of Things*
- IEEE - *Consumer Electronics Magazine*
- IEEE - *Transactions on Quantum Engineering*
- IEEE - *Transactions on Services Computing*
- IEEE - *Network*
- Elsevier - *Transportation Research*
- Elsevier - *Internet of Things*
- Elsevier - *Future Generation Computer Systems*
- Springer - *Nature*

Peer Reviewing - Conferences

- ACM - *DAC Chips to Systems Conference (DAC)* 2025, 2026
- ACM - *Design Automation and Test in Europe (DATE)* 2024, 2025, 2026
- IEEE - *Quantum Computing and Engineering (QCE)* 2023, 2024, 2025, 2026
- IEEE - *International Conference on Consumer Electronics (ICCE)* 2022, 2023, 2024

Society Memberships

- ACM - member
- IEEE - member
  - Vice-Chair for CTSoc Quantum in Consumer Technology (2025-2026). Website here.

Open Source Projects

QuFI *Torino, IT*
Developer *June 2021 - Ongoing*

- Development of *QuFI - Quantum Fault Injector*. Flexible tool highlighting the susceptibility of qubits to various noise sources. Related paper accepted at *DSN 22*.

Quantum Synth *Schilthorn, CH*
Developer *Sept. 12-15, 2019*

- Development of *Quantum synth*, the Quantum Computer-based Synthesizer. The project won the Community Choice Award at IBM Qiskit Camp Europe 2019. Video description available *here*. Related paper accepted at *AM'20*.

Talks

Colorado State University - Computer Science Department Colloquium *Fort Collins, CO, US*
Dependable Quantum-Classical Systems Engineering - A holistic approach *Feb. 21st, 2025*

- Developing dependable hybrid High-Performance Computing–Quantum Computing (HPC-QC) systems through robust co-design and fault management strategies.

Qiskit Fall Fest *UniNa, Naples*
Dependability Challenges in HPC-QC Integration *Nov. 11th, 2024*

- Talk on dependability challenges in integrating quantum computing systems within classical supercomputing infrastructures.

Washington DC Quantum Computing Meetup *Online*
Quantum Reliability: Circuit Susceptibility, Faults, and Integration Issues *May 12th, 2024*

- The talk addressed reliability challenges in quantum computing systems and integration challenges with traditional computing infrastructure, ultimately working toward improving the reliability and practical implementation of quantum computing technology. Video available *here*.

University of Illinois, Urbana-Champaign *Urbana-Champaign, IL, US*
Noisy Qubits: reliability issues in the quantum era *Nov. 29th, 2023*

- Presentation on noise-related issues in Quantum Computing and description of tools to analyze them. Presentation given at the attention of *Prof. Ravishankar K. Iyer* and the DEPEND research group.

University of Illinois, Chicago *Chicago, IL, US*
Noisy Qubits: reliability issues in the quantum era *Nov. 14th, 2023*

- Presentation on noise-related issues in Quantum Computing and description of tools to analyze them. Presentation given at the attention of the Quantum Information Science Society at UIC.

DSN23 *Porto, PT*
Quantum Computing Reliability: Problems, Tools, and Potential Solutions *June 27th, 2023*

- Tutorial on Quantum Computing Fault Injection. Co-organized with my colleagues Prof. Paolo Rech (UniTn), Prof. Devesh Tiwari (Northeastern U.), Dr. Emanuele Dri (PoliTo), Prof. Qiang Guan and Dr. Betis Baheri (Kent State U.).

INFN (National Institute for Nuclear Physics), La Sapienza University *Rome, IT*
Understanding transient fault propagation in superconducting quantum circuits *Nov. 21st, 2022*

- Seminar on Quantum Computing Fault Injection. Presentation given at the attention of *Dr. Laura Cardani* and her research group.

Technology Biennial (Biennale Tecnologia) - Politecnico di Torino *Torino, IT*
Quantum Computing Demo *Nov. 10-13, 2022*

- Booth activity dedicated to Quantum Computing, presentation of practical use cases for the financial domain. In collaboration with *Intesa Sanpaolo S.p.A.*.

NASA Goddard Space Flight Center *Greenbelt, MD, US*
Understanding transient fault propagation in superconducting quantum circuits *June 30th, 2022*

- Seminar on Quantum Computing Fault Injection. Presentation given at the attention of *Dr. Michael Campola* and his collaborators.

New York University, Center for Quantum Phenomena *NYC, NY, US*
Understanding transient fault propagation in superconducting quantum circuits *June 23rd, 2022*

- Seminar on Quantum Computing Fault Injection. Presentation given at the attention of *Prof. Javad Shabani* and his research group.

Technology Festival - Politecnico di Torino - Open Campus Event *Torino, IT*
Quantum Computing Demo *Nov. 7-10, 2019*

- Booth activity dedicated to Quantum Computing, presentation of *Quantum synth*, the Quantum Computer-based Synthesizer.

Politecnico di Torino *Torino, IT*
Quantum Computing Perspectives *June 20, 2019*

- Presentation on practical applications of Quantum Computing. Presentation given at the attention of *Prof. Matteo Sonza Reorda* and the CAD&Reliability Research Group

Teaching

PhD Degree *University of Naples Federico II, IT*
Instructor *Oct. 2024*

- Principles of IoE (PhD Course in Computational Intelligence - **English** Language)
  - 16 hours. AY 2024/2025.

Master Degree *University of Naples Federico II, IT*
Instructor *Mar. 2024 - July 2024*

- Quantum Computation Mod II - Architectures and High performance (Physics Major - **English** Language)
  - AY 2023/2024.

Master Degree *Politecnico di Torino, IT*
Teaching Assistant *Oct. 2019 - Oct. 2020*

- Computer Architectures (CS+EE major - **English** Language)
  - AY 2019/2020. Teaching reviews: 3.26/4.

Bachelor Degree *University of Naples Federico II, IT*
Instructor *Mar. 2025 - Sept. 2026*

- Computer Organization (Biomedical + Electronics + Telecommunications Engineering major - **Italian** Language).
- Foundations of Computer Architectures (Mechatronics major - **Italian** Language)

Bachelor Degree *Politecnico di Torino, IT*
Teaching Assistant *Oct. 2018 - Dec. 2023*

- Introduction to Computer Science (all majors - **Italian** Language).
  - AY 2018/2019. Teaching reviews: 3.19/4.
- Algorithms and Programming (EE major - **Italian** Language)
  - AY 2020/2021. Teaching reviews: 3.31/4.
  - AY 2021/2022. Teaching reviews: 3.38/4.
  - AY 2022/2023. Teaching reviews: 3.204/4.
- Programming Techniques (CS major - **English** Language)
  - AY 2020/2021. Teaching reviews: 3.225/4.
  - AY 2021/2022. Teaching reviews: 3.31/4.
  - AY 2022/2023. Teaching reviews: 3.056/4.

2nd **level Master Course on *Quantum Communication and Computing*** *Politecnico di Torino, IT*
Lecturer *Jan. 2023*

- Module on Quantum Annealing. (**Italian** Language).

Master Course - New Technologies 4.0 (for Leonardo S.p.A.) *Politecnico di Torino, IT*
Lecturer *July 2020*

- Lesson on Radio Frequency IDentification (RFID), A.Y. 2019/2020. (**Italian** Language).

Samsung Innovation Camp (for Samsung Italy) *Torino, IT*
Lecturer *Nov. - Dic. 2022*

- 10 hours course on Internet of Things. (**Italian** Language).

High School - Turin area *Politecnico di Torino, IT*
Lecturer *Apr. 2022 - Dec. 2023*

- *Computer Engineering is(/and) Creativity.* Course to foster female enrollment in computer engineering degree. (**Italian** Language).

General Information

Mother tongue Italian
English Language Certification TOEFL 101/120
Citizenship Italian